

Week 1 Guided Self-Study

- Computer Architecture 101 and Its Future
 - David Patterson (UC Berkeley & Google)
- Watch: the first 40 minutes
- Reflect: use the study questions on the next slide
- Focus: architectural concepts and tradeoffs, rather than product specifications

Study Questions

1. Why did the end of Dennard scaling change how architects improve performance?
2. How do memory bandwidth and capacity each limit the benefits of faster processors?
3. Why can domain-specific architectures achieve higher energy efficiency, and what do they sacrifice?
4. What does the Roofline model's ridge point represent, and how does it shift if peak compute throughput doubles while memory bandwidth stays unchanged?
5. What problems do chiplets and HBM address, and what is one tradeoff or limitation of each?